

S&S QUIZ- 8 SOLUTION

Sol(1)÷ Given causal LTI system's difference eqⁿ,
$$y[n] + \frac{1}{2}y[n-1] = x[n] \quad \text{--- (1)}$$

(a) As we know, $x[n] \xrightleftharpoons[\text{IDFT}]{\text{DFT}} X(e^{j\omega})$
where, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) \cdot e^{j\omega n} \cdot d\omega \quad \text{--- (2)}$$

Replacing $n \rightarrow (n-1)$ in eqⁿ(2)'s both side, we get -

$$x[n-1] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) \cdot e^{j\omega(n-1)} \cdot d\omega$$

$$x[n-1] = \frac{1}{2\pi} \int_{2\pi} e^{-j\omega} \cdot X(e^{j\omega}) \cdot e^{j\omega n} \cdot d\omega$$

$$\Rightarrow x[n-1] \xrightleftharpoons[\text{IDFT}]{\text{DFT}} e^{-j\omega} \cdot X(e^{j\omega}) \quad \text{--- (3) (1 mark)}$$

Taking DFT of both side of eqⁿ(1) with the help of eqⁿ(3),

$$Y(e^{j\omega}) + \frac{1}{2} e^{-j\omega} \cdot Y(e^{j\omega}) = X(e^{j\omega})$$

$$Y(e^{j\omega}) \left[1 + \frac{1}{2} e^{-j\omega} \right] = X(e^{j\omega})$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega} \right)}$$

$$\therefore H(e^{j\omega}) = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega} \right)} \quad \rightarrow (2 \text{ Marks})$$

(b) Given input, $x[n] = \left(-\frac{1}{2}\right)^n u[n] \quad \text{--- (4)}$

$$\text{As we know, } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n} \iff x[n]$$

$$\begin{aligned}
 x(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} \left(-\frac{1}{2}\right)^n u[n] e^{-j\omega n} \\
 &= \sum_{n=0}^{\infty} \left(-\frac{1}{2} e^{-j\omega}\right)^n \\
 &= \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \quad \text{--- (5)} \\
 &\quad \rightarrow (2 \text{ Marks})
 \end{aligned}$$

(C) Given input DFT, $x(e^{j\omega}) = 1 + 2e^{-j3\omega}$ --- (7)

By part (a), we know—

$$\begin{aligned}
 \frac{Y(e^{j\omega})}{X(e^{j\omega})} &= \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \\
 \therefore Y(e^{j\omega}) &= \frac{(1 + 2e^{-j3\omega})}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \\
 &= \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} + \frac{2e^{-j3\omega}}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \quad \text{--- (8)} \\
 &\quad \rightarrow (1 \text{ Marks})
 \end{aligned}$$

Taking IDFT of eqⁿ(8) with the help of eqⁿ(3) & eqⁿ(5),

$$y[n] = \left(-\frac{1}{2}\right)^n u[n] + 2 \left(-\frac{1}{2}\right)^{n-3} u[n-3]$$

→ (2 Marks)

Part (b) remaining part:

By part (a), we know -

$$\frac{X(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)}$$

$$Y(e^{j\omega}) = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)^2} \quad \text{--- (6)}$$

By eqⁿ(5), we get -

As we know that -

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$$

$$\frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$j \frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$\begin{cases} a^n u[n] \xrightarrow[\text{IDFT}]{\text{DFT}} \frac{1}{(1 - a e^{-j\omega})} \\ n a^n u[n] \xrightarrow[\text{IDFT}]{\text{DFT}} j \frac{d}{d\omega} \left[\frac{1}{1 - a e^{-j\omega}} \right] \\ (n+1) a^n u[n+1] \xrightarrow[\text{IDFT}]{\text{DFT}} \frac{1}{(1 - a e^{-j\omega})^2} \end{cases} \quad \text{--- (A)}$$

→ (1 Marks)

Taking IDFT of eqⁿ(6) by using eqⁿ(7), we get -

$$y[n] = (n+1) \left(-\frac{1}{2}\right)^n u[n+1]$$

→ (1 Marks)

Taking IDFT of eqⁿ(8) with the help of eqⁿ(3) & eqⁿ(5),

$$y[n] = \left(-\frac{1}{2}\right)^n u[n] + 2 \left(-\frac{1}{2}\right)^{n-3} u[n-3]$$

→ (2 Marks)

Part (b) remaining part:

By part (a), we know -

$$\frac{X(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)}$$

$$Y(e^{j\omega}) = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)^2} \quad \text{--- (6)}$$

By eqⁿ(5), we get -

As we know that -

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$$

$$\frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$j \frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$\begin{cases} a^n u[n] \xrightarrow[\text{IDFT}]{\text{DFT}} \frac{1}{(1 - a e^{-j\omega})} \\ n a^n u[n] \xrightarrow[\text{IDFT}]{\text{DFT}} j \frac{d}{d\omega} \left[\frac{1}{1 - a e^{-j\omega}} \right] \\ (n+1) a^n u[n] \xrightarrow[\text{IDFT}]{\text{DFT}} \frac{1}{(1 - a e^{-j\omega})^2} \end{cases} \quad \text{--- (A)}$$

→ (1 Marks)

Taking IDFT of eqⁿ(6) by using eqⁿ(7), we get -

$$y[n] = (n+1) \left(-\frac{1}{2}\right)^n u[n]$$

→ (1 Marks)